Predicting Real Estate Prices

A Statistical and Machine Learning Study of Housing Prices per Unit Area

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Abstract

This project studies how housing characteristics, accessibility, amenities, transaction timing, and spatial location are associated with house prices per unit area, and compares interpretable regression models with more flexible machine-learning methods in out-of-sample prediction. Using a dataset with 414 residential transactions, we first show that a naive additive linear model is misspecified: exploratory plots suggest nonlinear effects of MRT accessibility and house age, while baseline OLS diagnostics strongly reject the simple linear functional form. We then compare baseline OLS, feature-engineered OLS, regularized linear models, spline regression, tree-based ensembles, and a neural-network benchmark. The main interpretable model is an engineered OLS specification with log(price) as the response and nonlinear terms for age and MRT distance. This model achieves $R^2 = 0.741$ and adjusted $R^2 = 0.736$, and indicates that transaction date is positively associated with price, distance to the nearest MRT station is strongly negatively associated with price, and house age has a nonlinear association with price. In repeated cross-validation, random forest performs best overall, while feature-engineered OLS substantially improves on baseline OLS. Additional diagnostics show that feature engineering largely fixes the strongest functional-form problems in baseline OLS, and that modeling log(price) yields more stable residual behavior than modeling price directly. Overall, the results suggest that accessibility and location are the dominant drivers of price variation, while flexible nonlinear models improve predictive accuracy without overturning the main substantive conclusions.

1 Introduction

Housing prices are jointly shaped by structural, locational, and neighborhood characteristics, consistent with the classical hedonic pricing framework in which real-estate values reflect the implicit prices of observed housing attributes [1]. In urban real-estate markets, accessibility to transportation, nearby amenities, housing age, and broader spatial context often all matter for local valuation [2, 3]. At the same time, these effects need not be linear, and spatial dependence can complicate both interpretation and prediction [4, 5]. Recent work also shows

that machine-learning methods can outperform simpler benchmarks in property-price prediction when nonlinearities and interactions are important [6]. This dataset is statistically interesting for three reasons. First, the relation between price and MRT accessibility is plausibly nonlinear. Second, house age may exhibit curvature rather than a purely linear effect. Third, spatial coordinates likely proxy for unobserved neighborhood quality, which creates a tension between parsimonious interpretation and predictive flexibility. For these reasons, the project is designed around both explanation and prediction. The analysis addresses the following questions: (i) what are the main statistical associations between housing price and the observed predictors; (ii) how much feature engineering and nonlinear modeling improve over a naive linear benchmark; and (iii) whether the main conclusions are robust to response transformation and influential observations. The rest of the paper proceeds as follows. Section 2 describes the data and exploratory analysis. Section 3 outlines the modeling strategy. Section 4 reports the main results. Section 5 presents diagnostics and robustness checks. Section 6 concludes. Supplementary plots, including machine learning interpretability and spatial error diagnostics, are placed in the appendix so that the main text remains within the 10-page limit.

2 Data and Exploratory Analysis

The dataset contains $n = 414$ residential transactions and corresponds to the real-estate valuation dataset introduced by Yeh and Hsu [7]. The response variable is house price per unit area. The predictors are transaction date, house age, distance to the nearest MRT station, number of nearby convenience stores, latitude, and longitude. Because the sample is observational rather than experimental, all results should be interpreted as conditional associations rather than causal effects.

Table 1 summarizes the variables used in the analysis. To motivate later modeling choices, we construct three engineered variables: log(dist_to_mrt) to reduce the strong right-skew of MRT distance, age^2 to capture curvature in the age-price relationship, and a latitude–longitude interaction to allow for richer spatial structure. The full scatter matrix is reported in Appendix Figure 2.

Table 1: Variable definitions used in the analysis

Variable	**Description**
price	House price per unit area (original response variable)
log_price	Log-transformed house price per unit area
transaction_date	Transaction timing variable
house_age	Age of the house
dist_to_mrt	Distance to the nearest MRT station
log_dist	log(dist_to_mrt)
stores	Number of nearby convenience stores
latitude	Geographic latitude
longitude	Geographic longitude
age_sq	Squared house age, $(\text{house_age})^2$
lat_lon	Interaction term, latitude $\times$ longitude

The EDA suggests four main patterns. First, price decreases strongly with distance to MRT, but the relationship is visibly curved rather than linear; the slope is steep near low distances and flattens for larger distances. Second, plotting price against log(distance) is noticeably closer to linear than plotting price against raw distance, which motivates the log transform. Third, house age shows a non-monotone pattern, supporting the inclusion of a quadratic term. Fourth, latitude and longitude display clear spatial clustering, suggesting that location contains residual information not fully captured by accessibility and amenities alone. These empirical patterns imply three modeling priorities: baseline OLS should be treated only as a reference model; any serious linear specification should allow nonlinear effects for MRT distance and house age; and it is useful to compare interpretable regression against more flexible nonlinear methods.

3 Methodology

3.1 Model families

We compare several model classes designed to answer slightly different questions.

Baseline OLS. The benchmark linear model uses the original predictors in an additive form:

$$\begin{aligned}\text{price}_i = \beta_0 + \beta_1\text{transaction_date}_i + \beta_2\text{house_age}_i + \beta_3\text{dist_to_mrt}_i \\ + \beta_4\text{stores}_i + \beta_5\text{latitude}_i + \beta_6\text{longitude}_i + \varepsilon_i\end{aligned} \tag{1}$$

This model is intentionally simple and serves as a diagnostic reference point.

Engineered OLS. The main interpretable specification incorporates nonlinear terms motivated by EDA:

$$\begin{aligned}\log(\text{price}_i) = \beta_0 &+ \beta_1 \text{transaction_date}_i + \beta_2 \text{house_age}_i + \beta_3 \text{house_age}_i^2 \\ &+ \beta_4 \log(\text{dist_to_mrt}_i) + \beta_5 \text{stores}_i + \beta_6 \text{latitude}_i \\ &+ \beta_7 \text{longitude}_i + \beta_8 (\text{latitude}_i \times \text{longitude}_i) + \varepsilon_i\end{aligned} \tag{2}$$

This is the main model used for coefficient-based interpretation.

Other predictive models. To assess whether additional flexibility improves generalization, we also fit ridge and lasso on the engineered features, a spline-based linear model, gradient boosting, random forest, and an MLP neural-network benchmark, following the broader literature that compares statistical and machine-learning approaches for property valuation [6].

3.2 Evaluation strategy

Model comparison is based on repeated cross-validation using RMSE, MAE, and R^2. For the specific raw-price versus log-price OLS comparison, we report RMSE on the *original price scale*; for the log-response model, predictions are exponentiated before computing RMSE. This makes the two RMSE values directly comparable. By contrast, RMSE computed on the log scale itself is not directly comparable with raw-price RMSE because the units differ. For regression diagnostics, we examine residual plots, Q–Q plots, scale-location plots, influence measures, Breusch–Pagan tests for heteroscedasticity, and Ramsey RESET tests for functional-form misspecification. The central logic is diagnostic-to-remedy: baseline OLS is first used to identify the main problems, and engineered OLS is then evaluated in terms of how far those problems are reduced.

4 Results

4.1 Main interpretable model

Table 2 reports the final engineered OLS model with log(price) as the response. The model explains a substantial share of variation in the data, with $R^2 = 0.741$ and adjusted $R^2 = 0.736$.

Table 2: Final engineered OLS model for log(price)

Variable	**Coef.**	**Std. Err.**	**t**	**p-value**
Intercept	397100.0000	176700.0000	2.250	0.025
transaction_date	0.1700	0.0360	4.768	< 0.001
house_age	-0.0225	0.0040	-6.413	< 0.001
age_sq	0.0004	0.000085	4.813	< 0.001
log_dist	-0.1522	0.0150	-9.928	< 0.001
stores	0.0090	0.0050	1.841	0.066
latitude	-15940.0000	7072.1030	-2.254	0.025
longitude	-3272.5976	1452.3690	-2.253	0.025
lat_lon	131.2335	58.1930	2.255	0.025
Observations = 414, $R^2 = 0.7413$, Adjusted $R^2 = 0.7362$, AIC = -142.31, BIC = -106.08				

Three findings stand out. First, transaction date is positively associated with price, indicating an upward trend over the sample period. Second, log(distance to MRT) is strongly negative and highly significant, showing that accessibility is one of the most important determinants of value in this dataset. Third, house age enters with both a negative linear term and a positive quadratic term, implying a nonlinear age effect rather than a purely linear depreciation pattern. The coefficient on convenience stores is positive but statistically less stable than those for date, distance, and age, suggesting that some of its predictive information overlaps with broader locational covariates. The spatial terms are jointly informative, but their individual coefficients should be interpreted cautiously because latitude, longitude, and their interaction create substantial collinearity when entered without centering. Substantively, the relevant conclusion is collective: even after controlling for accessibility and amenities, residual location value still matters.

4.2 Cross-validated model comparison

Table 3 summarizes predictive performance across model classes. Two patterns are immediate. First, baseline OLS is the weakest of the main competitors, which confirms that naive linearity is too restrictive. Second, feature engineering closes much of the gap between simple linear regression and more flexible machine-learning methods. Random forest achieves the best average predictive performance overall, while engineered OLS, spline regression, and boosting all perform substantially better than the baseline linear model.

Table 3: Cross-validated predictive performance comparison

Model	**RMSE Mean**	**RMSE SD**	**MAE Mean**	**MAE SD**	R^2 **Mean**	R^2 **SD**
Random forest	7.4012	1.5923	4.8421	0.5090	0.6984	0.0710
Gradient boosting	7.7002	1.5484	5.0942	0.4860	0.6744	0.0642
Spline linear	7.7651	1.5421	5.1060	0.5568	0.6674	0.0741
OLS engineered	7.7945	1.5433	5.3152	0.5381	0.6659	0.0673
Lasso engineered	7.9351	1.5634	5.3231	0.6234	0.6531	0.0730
Ridge engineered	7.9396	1.5623	5.3246	0.6241	0.6526	0.0731
MLP neural net	8.6961	1.9648	6.0990	1.4119	0.5767	0.1507
OLS baseline	8.8782	1.6293	6.3236	0.7639	0.5657	0.0797

The neural-network benchmark does not outperform the best tree-based or engineered linear models. Given the modest sample size and tabular structure of the data, this is not surprising and is broadly consistent with recent evidence that flexible machine-learning methods can be highly competitive in property-price prediction when nonlinearities and interactions are important [6]. To further understand the structure learned by these flexible nonlinear models, we examine feature importance and partial dependence in Appendix B.

5 Diagnostics and Robustness

5.1 Diagnostics for baseline OLS

The baseline OLS model is useful mainly because it reveals what a naive linear specification misses. Table 4 shows that baseline OLS has $R^2 = 0.582$, but the Ramsey RESET test strongly rejects the simple functional form ($p < 10^{-12}$). This is the clearest evidence that the additive linear benchmark is misspecified. By contrast, the Breusch–Pagan test is not significant at conventional levels, so the most serious issue is not classic heteroscedasticity but functional-form failure. The residual distribution is also markedly non-normal, with a Shapiro–Wilk p-value far below 0.001.

5.2 How far engineered OLS fixes them

The engineered models materially improve on the baseline. Moving from baseline OLS to engineered OLS on the raw price scale raises R^2 from 0.582 to 0.687 and, more importantly, makes the RESET test insignificant ($p = 0.481$), indicating that the strongest functional-form problem is largely resolved. This supports the EDA-based choice to use log(distance) and a quadratic age term. Moving from raw-price engineered OLS to the log-response engineered model improves fit further to $R^2 = 0.741$ and lowers the maximum Cook's distance from 0.235 to 0.095. In other words, the log-response formulation is not just a cosmetic transformation: it produces more stable residual behavior and reduces the influence of extreme observations.

Non-normal tails remain, so the transformed model is not "perfect," but it is clearly more defensible than either baseline OLS or raw-price engineered OLS.

Table 4: Diagnostic summary for baseline and engineered OLS models

Model	R^2	**RESET** p	**BP** p	**Shapiro** p	**Max Cook's D**
Baseline OLS (price)	0.582	1.66×10^{-13}	0.206	1.10×10^{-17}	0.195
Engineered OLS (price)	0.687	0.481	0.283	4.04×10^{-20}	0.235
Engineered OLS (log price)	0.741	0.010	0.469	1.80×10^{-15}	0.095

A related issue is multicollinearity in the spatial terms. When latitude, longitude, and their interaction are entered without centering, their VIF values are extremely large. Recentering latitude and longitude before forming the interaction leaves model fit essentially unchanged but reduces those VIF values to ordinary levels. Therefore, the multicollinearity problem is best viewed as a scaling issue in the uncentered interaction rather than as evidence that spatial information should be removed from the model. Further error analysis, including out-of-fold predictions and mapping of spatial residuals, is provided in Appendix C.

5.3 Raw-price versus log-price RMSE

Because RMSE depends on the response scale, raw-price and log-price OLS can be compared fairly only after putting predictions on the same scale. We therefore compute cross-validated RMSE for both models on the original price scale. The raw-price engineered OLS model achieves a mean RMSE of 7.45, while the log-price engineered OLS model, after back-transformation, achieves a mean RMSE of 7.56. The gap is small, so neither model dominates decisively on original-scale prediction alone. However, the log-response model has clearly better diagnostics and is less sensitive to influential observations, which is why it remains the preferred inferential specification.

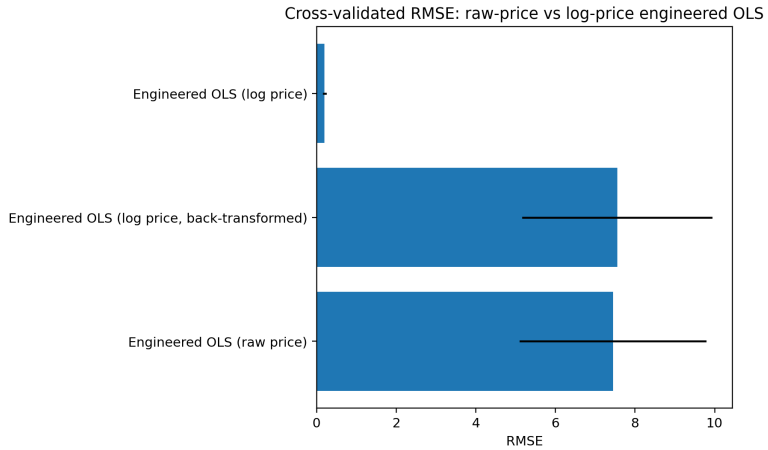

Figure 1: Cross-validated RMSE for engineered OLS on the original price scale. The two models are very close, so the preference for the log-response model is driven mainly by diagnostics and stability rather than by a large RMSE gain.

6 Conclusion

This project finds that housing value in the dataset is driven primarily by transportation accessibility and spatial context, with house age entering nonlinearly and convenience stores providing weaker conditional evidence once broader locational information is included. From a modeling perspective, the analysis shows that feature engineering is not a minor adjustment: it removes the most severe functional-form problems in the naive linear model and substantially improves prediction. Tree-based methods perform best in out-of-sample prediction, but the engineered log-response OLS model offers the clearest coefficient-based interpretation and the most defensible diagnostics among the linear specifications. For this reason, the report supports a dual conclusion: random forest is the best predictive model, while engineered log-response OLS is the preferred inferential model. Future work could extend the analysis using richer neighborhood covariates and explicit spatial modeling, since recent work suggests that accounting more directly for spatial autocorrelation can matter for both performance evaluation and house-price prediction [4, 5, 8].

A Supplementary Plots and Notes

The appendix contains supplementary figures that are informative but not essential for the main narrative. These materials are not counted toward the 10-page main-text limit.

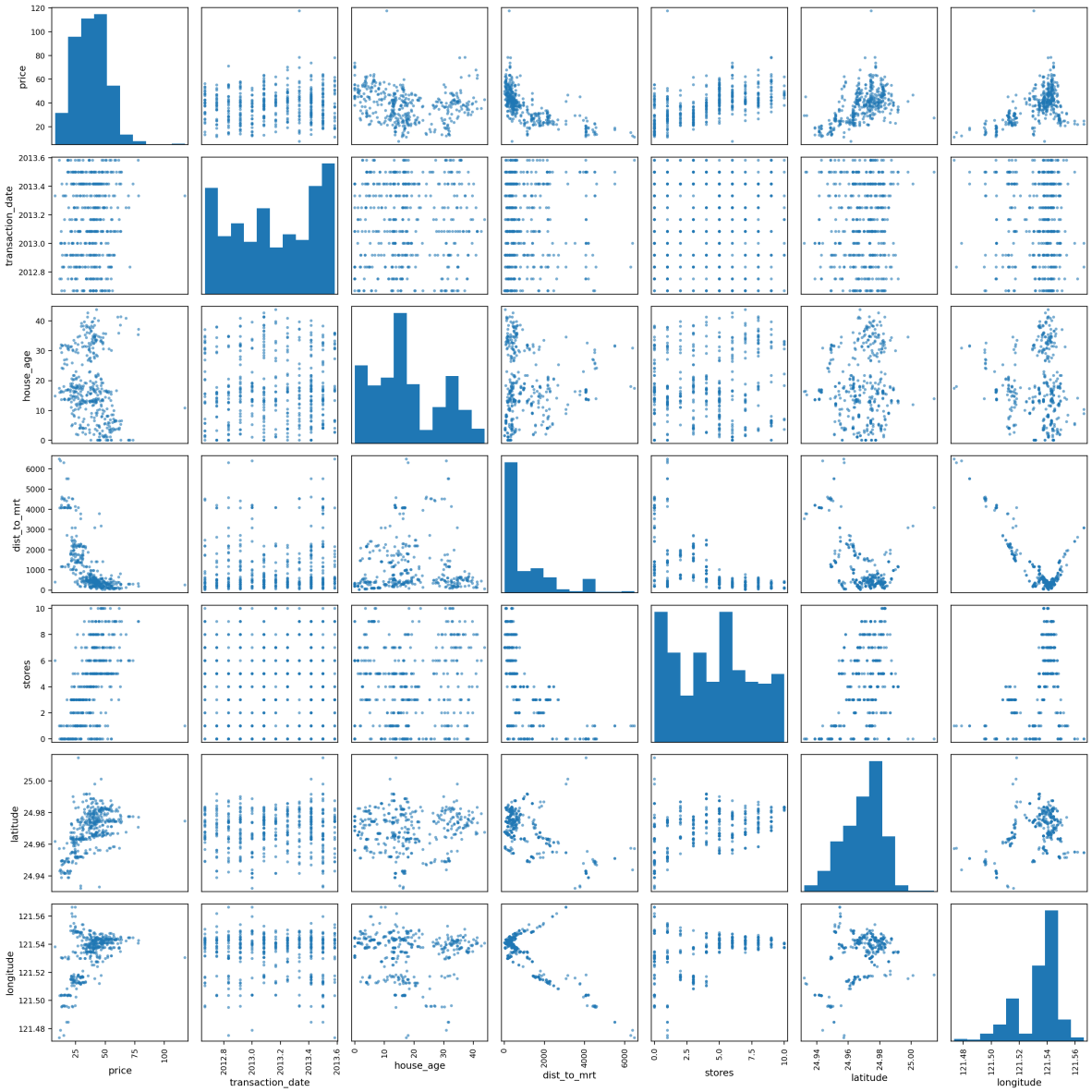

Figure 2: Scatter matrix of all observed variables used in the exploratory analysis. The matrix shows the strong negative and nonlinear relationship between price and MRT distance, visible curvature in the age-price relationship, and clear spatial clustering in latitude and longitude.

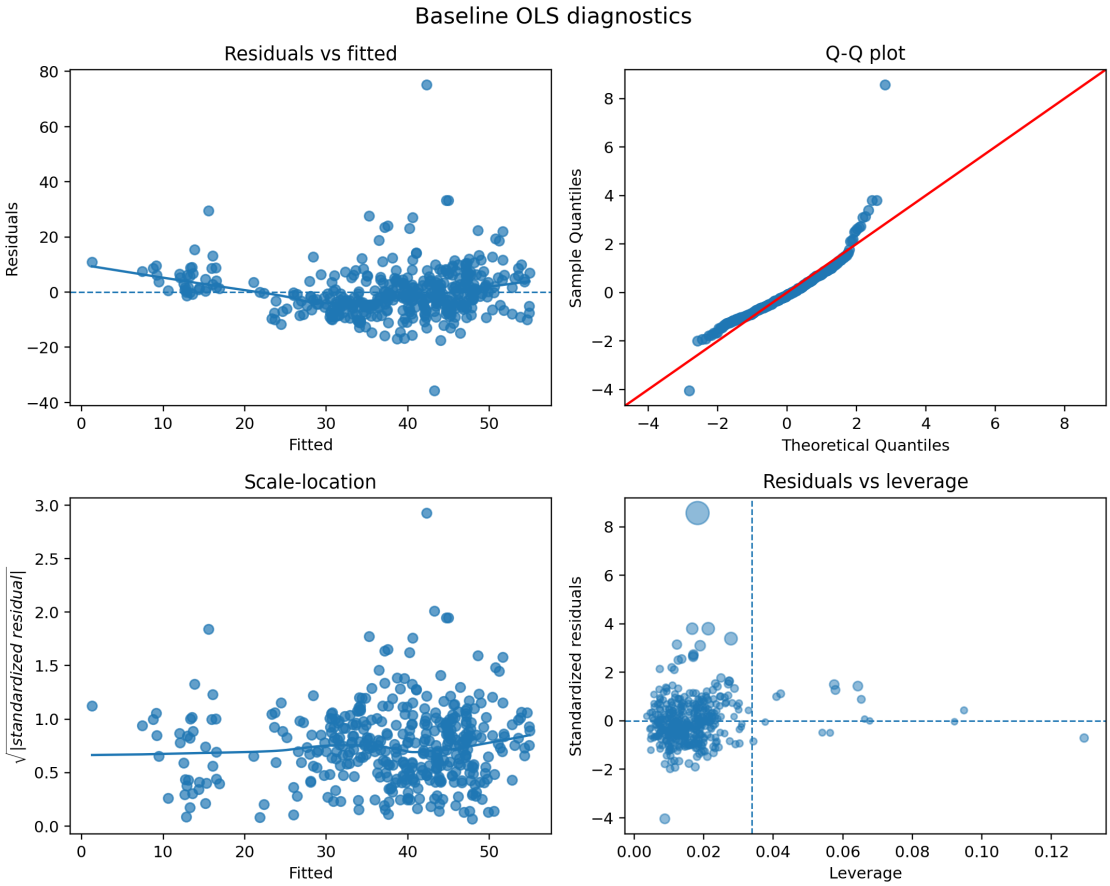

Figure 3: Diagnostic plots for baseline OLS. The residual-versus-fitted panel shows systematic curvature, consistent with the highly significant RESET test in the main text.

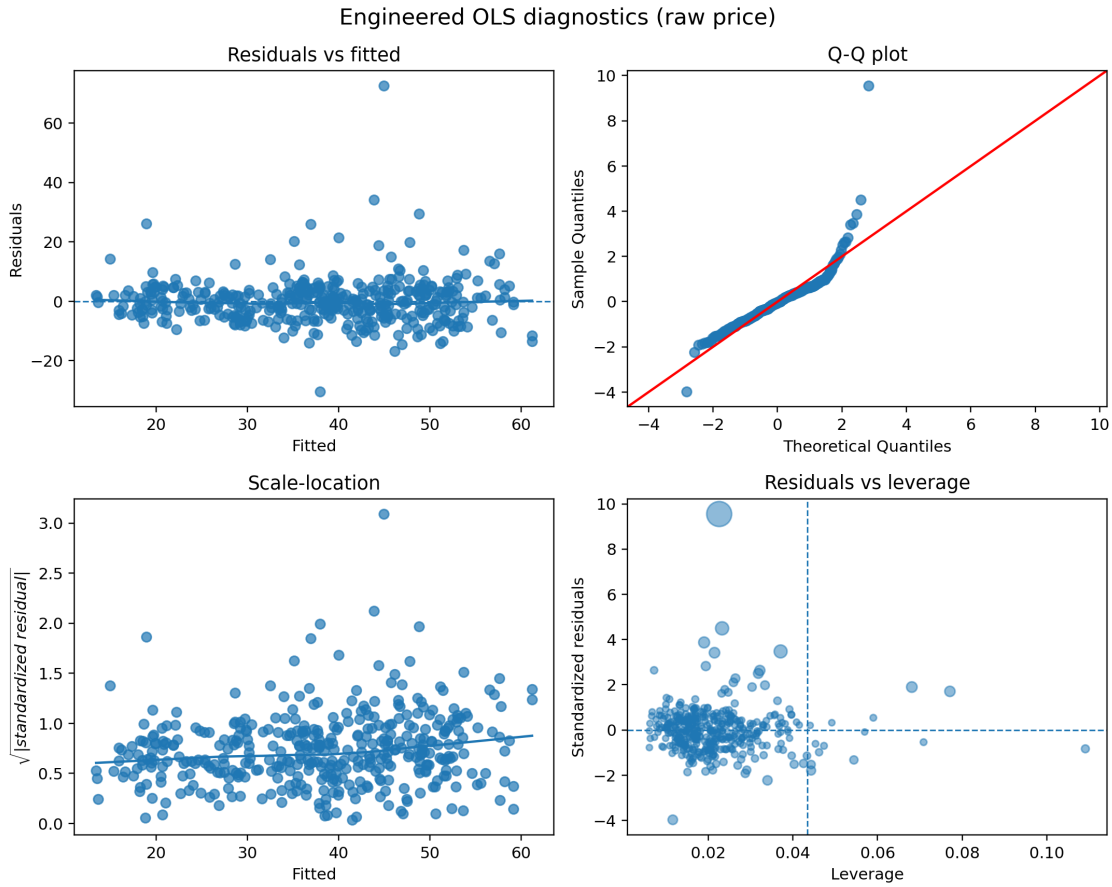

Figure 4: Diagnostic plots for engineered OLS with raw price as the response. Functional-form fit improves relative to baseline OLS, but influential observations remain more prominent on the raw scale.

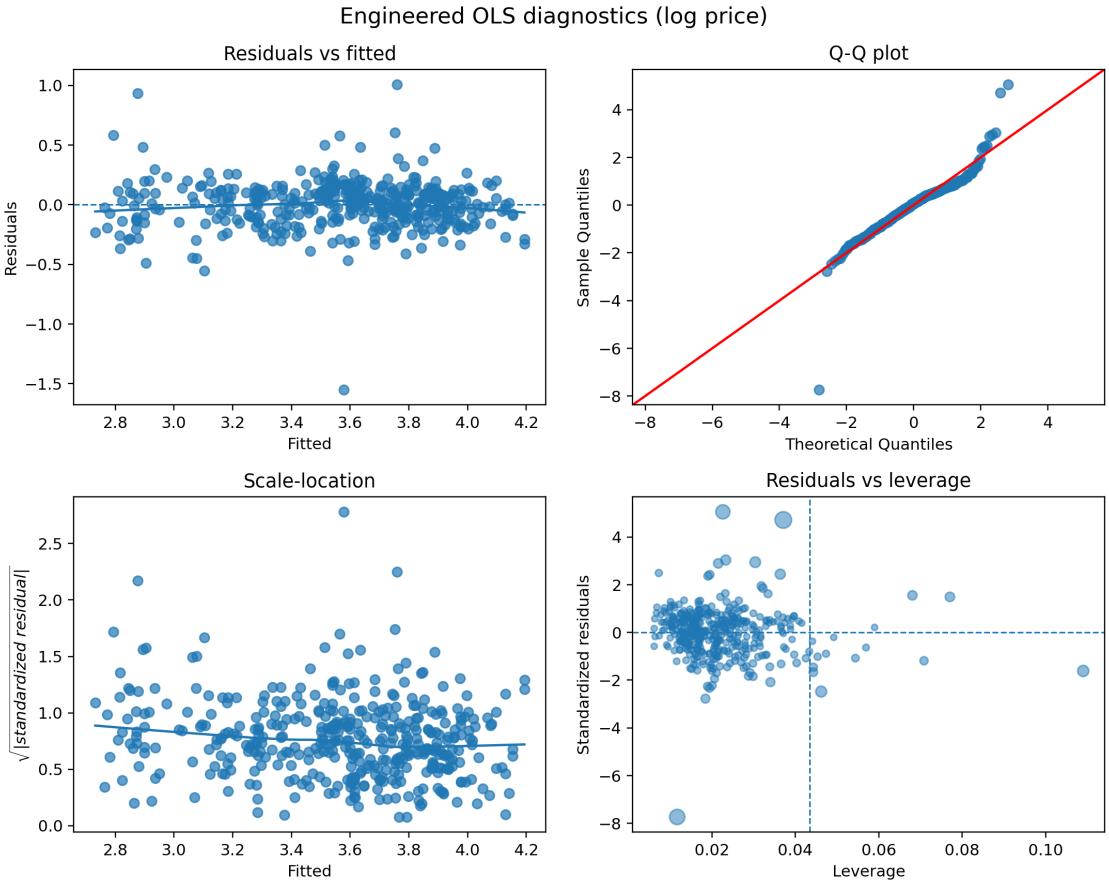

Figure 5: Diagnostic plots for engineered OLS with log(price) as the response. The log-response formulation yields more stable residual behavior and lower influence from extreme observations.

B Machine Learning Interpretability

To better understand the structure learned by the flexible nonlinear models, we examine feature importance and partial dependence from the gradient boosting model. Figure 6 shows that accessibility and location-related variables are among the most important predictors, consistent with the regression results.

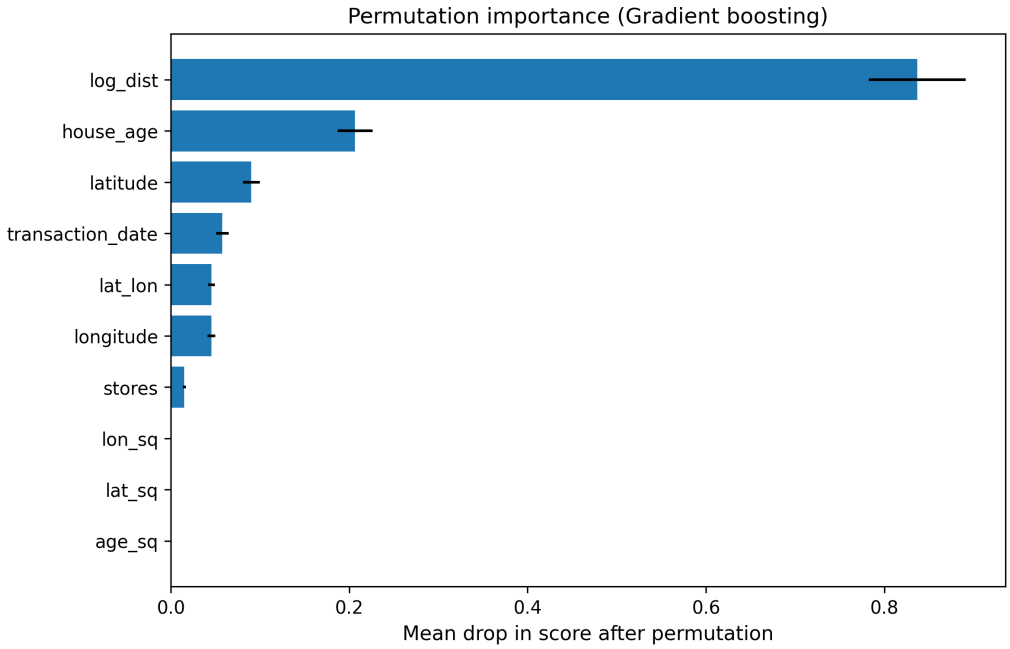

Figure 6: Permutation importance from the gradient boosting model. Accessibility and spatial variables provide the largest contributions to predictive performance.

The partial dependence analysis in Figure 7 indicates a pronounced negative relationship between `log_dist` and predicted price, especially over lower-to-middle ranges of the variable. House age also shows a non-monotone pattern, supporting the inclusion of nonlinear structure in the engineered regression models. By contrast, the effect of `stores` appears comparatively weak and flat after accounting for the other predictors.

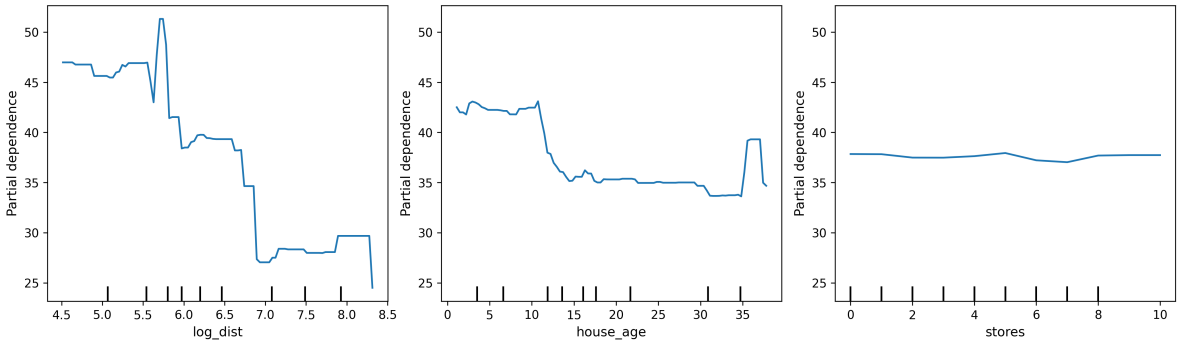

Figure 7: Partial dependence plots from the gradient boosting model. The results support nonlinear effects for MRT distance and house age, while the marginal contribution of convenience stores is relatively modest after conditioning on other predictors.

C Error Analysis and Spatial Residuals

We examine out-of-fold (OOF) residual behavior for the best predictive model, random forest. Figure 8 suggests that the model fits the bulk of the data reasonably well, but it tends to underpredict some of the highest-priced observations. This is a common pattern in ensemble models trained on moderately sized tabular datasets, where extreme observations may be smoothed toward the center.

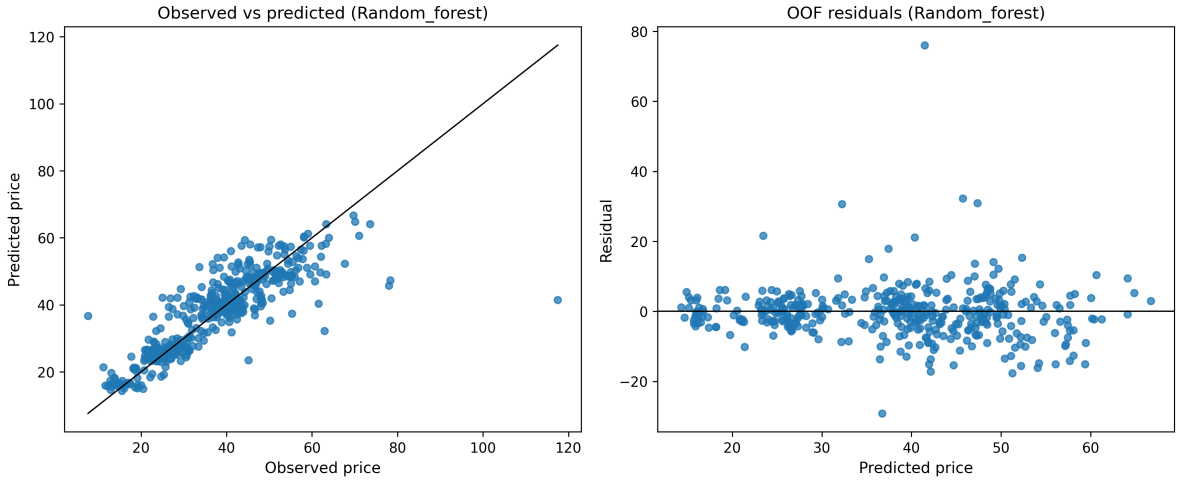

Figure 8: Observed-versus-predicted and residual plots for out-of-fold predictions from the random forest model. The fit is strong overall, but some expensive properties are underpredicted.

Figure 9 maps prediction error spatially. The residual pattern suggests that even the best-performing predictive models may miss some neighborhood-level or micro-location factors that are not fully captured by the observed variables. This reinforces the substantive importance of location in housing-price formation.

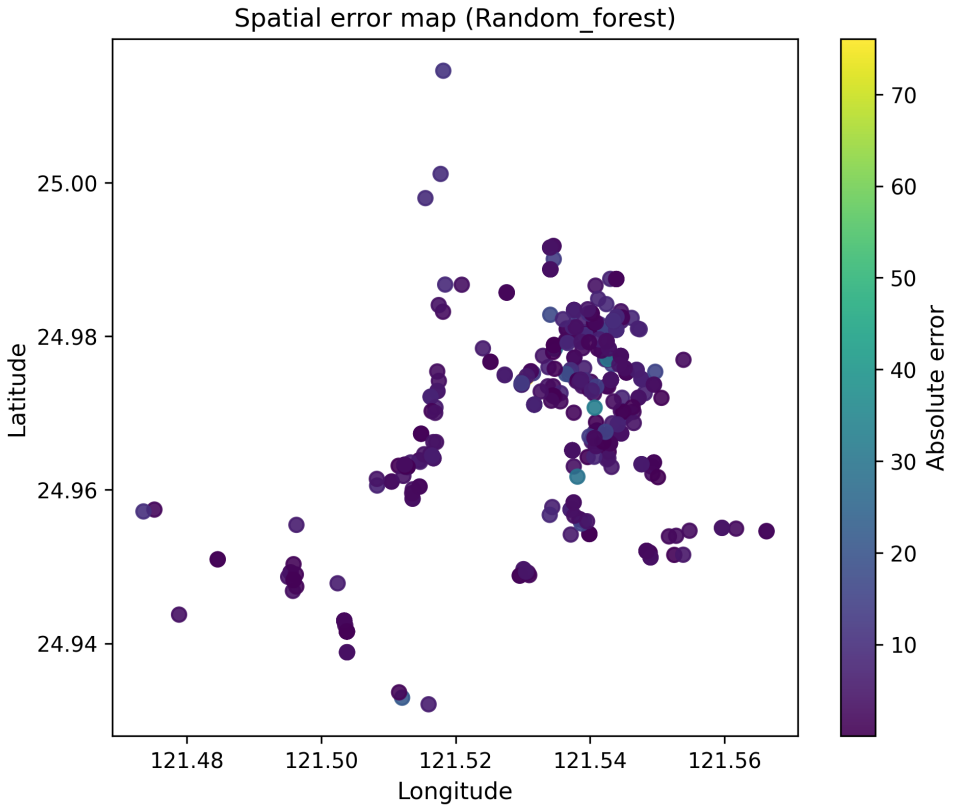

Figure 9: Spatial distribution of model residuals. Remaining spatial structure suggests that some neighborhood-specific value is not fully captured by the observed predictors.

D Code Availability

The empirical analysis was implemented in Python using `pandas`, `statsmodels`, and `scikit-learn`. Due to space limitations, the complete code used for this project has been uploaded to a GitHub repository, available at: https://github.com/wxr6666666/DSC-241-Project/tree/main.

E LLM Disclosure

Large language model (LLM) tools were used only in a limited supporting role during the preparation of this project. In particular, LLM assistance was used for partial planning of the project workflow and report structure, including helping organize the paper into the sequence of introduction, exploratory data analysis, methodology, results, diagnostics, conclusion, and appendix, and helping frame the distinction between interpretable statistical modeling and predictive machine-learning modeling.

LLM tools were also used for limited implementation support in coding. This included assistance with parts of the Python analysis workflow, such as drafting or revising code for

variable construction (for example, the log transformation of MRT distance, the quadratic age term, and the latitude–longitude interaction), repeated cross-validation pipelines, regression diagnostics, and figure generation. Some support was also used for checking code syntax, improving code organization, and troubleshooting parts of the implementation involving `pandas`, `statsmodels`, and `scikit-learn`. In addition, LLM assistance was used for minor writing and formatting tasks, such as improving wording, polishing transitions between sections, refining table and figure captions, and adjusting LaTeX layout for equations, tables, appendix materials, and overall presentation.

However, the LLM was not used as a substitute for the authors' own statistical reasoning or final judgment. All substantive analytical decisions were made by the authors, including the choice to compare baseline OLS, engineered OLS, regularized linear models, spline regression, gradient boosting, random forest, and an MLP benchmark; the decision to model both raw price and log-transformed price; the interpretation of diagnostics such as RESET, Breusch–Pagan, Shapiro–Wilk, and Cook's distance; and the final conclusions regarding prediction versus interpretation. All numerical results, tables, figures, and written interpretations were reviewed, checked, and revised by the authors, who take full responsibility for the accuracy, originality, and integrity of the final submission.

F Contribution Statement

Xurui Wang and Yuru Feng contributed equally to this project. Both authors jointly participated in all major stages of the assignment, including exploratory data analysis, feature engineering, model implementation, diagnostic analysis, predictive evaluation, interpretation of results, figure and table preparation, and final report writing.

Specifically, both authors worked on the empirical analysis of the housing dataset, including construction of engineered variables such as log(dist_to_mrt), house_age^2, and the latitude–longitude interaction term; estimation and comparison of baseline and engineered OLS models; implementation of regularized linear models, spline regression, gradient boosting, random forest, and MLP models; repeated cross-validation using RMSE, MAE, and R^2; and robustness checks comparing raw-price and log-price specifications. Both authors also jointly examined model diagnostics, including residual plots, Q–Q plots, scale-location plots, RESET tests, Breusch–Pagan tests, Shapiro–Wilk tests, influence diagnostics, and spatial residual patterns.

In addition, both authors contributed equally to drafting, revising, and formatting the written report, including the abstract, methodological exposition, interpretation of statistical output, appendix materials on machine-learning interpretability and spatial error analysis, and final presentation in LaTeX. The final submission therefore reflects equal intellectual and practical contributions from Xurui Wang and Yuru Feng.

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